

Project Proposal

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Student Number	2524641
Degree Pathway (course)	BSc. (Hons) Software Engineering (TopUp)
Supervisor Name	Mr. Vibhavi Artigala
Title of Project	TripMate: A Webapp for collaborative planning and managing trips in Sri Lanka
Abstract of the project	The proposed project, Tripmate, is a WebApp that will make it easier and more enjoyable for people to plan trips. The process of organizing a group trip with friends, couples, or family is often disorganized and inefficient in that different services are used to communicate and arrange a trip. The proposed project will eradicate this problem by providing a single platform where users can create, update and collaborate on trip plans in real time. A user can invite members with a link, where everyone in the group can suggest destinations, make routes and vote for their favourite. The WebApp is implemented using modern web technologies and free Application Programming Interfaces (API) for interactive mapping, weather forecasting and displaying popular travel videos for selected destinations. Natural Language Processing (NLP) models are presented to process user queries and produce a contextual travel information based on video, weather and mapping APIs to analyse and recommend destinations. Also, depending on the trip distance, the type of vehicle and the preferred lodging the system forecasts and estimates the costs of traveling, the overall cost, and the cost per head using Logistic Regression. Every user within this project will have a chance to develop different personalized trips through a secure user-authentication process and will ensure that all members are accordingly in sync within the group. The proposed project makes group travel planning more efficient, transparent and enjoyable with the combination of the above technologies.
Project deliverables	Software: A WebApp designed for group trip planning and management. Reports: - Project proposal - Contextual report - Thesis report - Design document - User manual
Description of your artefact	The proposed project is a WebApp that will make collaborative trip planning and management easier. The proposed system has features to manage a shared trip plan for a group of people. It has features of allowing group members to add destinations and a member vote for routes and hotels. To improve user experience and decision making, proposed project combines mapping, weather, and video APIs with the help of NLP models. The user-generated text queries will be interpreted using NLP and will automatically search the relevant destinations, weather conditions, and travel videos using the integrated APIs.

The proposed project also uses a Logistic Regression model, which is used to estimate intelligent travel costs and predict the total and the per-person expenses, depending on the distance of the trip, the type of vehicle, the efficiency of its fuel usage, the number of travelers, and the preferences of accommodation.

Other similar WebApps like TripIt, Google Travel, the red value, Trip advisor offers personal trip management but not collaborative planning. Thus, the proposed project is distinctive in offering a collaborative trip management facility for making group decisions (TripIt, 2025; Google Travel, 2025; The Red Value 2025; Trip Advisor 2025).

TripIt (no date) *TripIt: Travel Planner App*. Available at: <https://www.tripit.com> (Accessed: 28 October 2025).

Google Travel (no date) *Travel Planner by Google*. Available at: <https://www.google.com/travel> (Accessed: 28 October 2025).

The Japan map (no date) *The Red Value*. Available at: <https://www.theredvalue.com/map/> (Accessed: 28 October 2025).

Tripadvisor (No date): Over a billion reviews & contributions for hotels, attractions, restaurants, and more. Available at: <https://www.tripadvisor.com/> (Accessed: 28 October 2025).

Aim:

To create a smart, collaborative web-based trip planning tool that also uses Natural Language Processing and predictive modelling to help groups plan, organize, and manage trips.

Objectives:

- To design and implement an interactive web interface that supports group trip planning.
- To integrate mapping, weather, and video APIs for destination discovery and real-time information with the help of natural language process models.
- To develop a secure authentication system for managing user profiles.
- To implement cost estimation features for calculating total and per-person travel expenses using logistic regression.
- To establish an authenticated information system to use in providing individual access and group control.

List of Features:

Functional Features:

- Registration and login features for secure user login.
- Designing and organization of joint trip plans collaboratively.
- Invitation links for group collaboration.
- Voting mechanism for route and destination selection.
- The interactive map shows destinations and routes.
- Forecast information for each day during your journey.
- Integration of YouTube videos for destination information and trending destinations.
- Cost estimation depending on fuel costs, accommodation fees, and group spending.
- NLP based destination suggestion and keyword deciphering

	Non-Functional Features: • Cross browser response and user-friendly interface. • Collaborative synchronization of real-time data. • Ensuring data integrity and security through correct authentication and validation. • Scalability for future integration of additional destinations or services. • Maintainability with a modular structure and controlling versions via GitHub.																					
Risk analysis	<table><tr><th>Risk</th><th>Impact</th><th>Mitigation Actions</th></tr><tr><td>API dependency failure</td><td>High</td><td>Use API keys securely and put error handling for unsuccessful requests in place.</td></tr><tr><td>Server problems or downtime</td><td>High</td><td>Use a reliable host, do load testing, and make sure to do regular system maintenance.</td></tr><tr><td>Data inconsistency or loss</td><td>High</td><td>Perform regular database backups, apply data validation rules, and use secure transactions</td></tr><tr><td>Security vulnerabilities in user authentication</td><td>High</td><td>To protect against attacks, use strong password encryption, session handling and input sanitization.</td></tr><tr><td>High workload during group collaboration</td><td>Medium</td><td>Use role-based permissions and visual indicators to see who changed or updated data in real time.</td></tr><tr><td>Cross-browser compatibility issues</td><td>Low</td><td>Test the system on major browsers and use responsive designs.</td></tr></table>	Risk	Impact	Mitigation Actions	API dependency failure	High	Use API keys securely and put error handling for unsuccessful requests in place.	Server problems or downtime	High	Use a reliable host, do load testing, and make sure to do regular system maintenance.	Data inconsistency or loss	High	Perform regular database backups, apply data validation rules, and use secure transactions	Security vulnerabilities in user authentication	High	To protect against attacks, use strong password encryption, session handling and input sanitization.	High workload during group collaboration	Medium	Use role-based permissions and visual indicators to see who changed or updated data in real time.	Cross-browser compatibility issues	Low	Test the system on major browsers and use responsive designs.
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How does your project relate to your degree course and build upon the units/knowledge you have studied/acquired	The proposed project is closely aligned with the knowledge acquired through the degree path. In modules like Software for Enterprise and Agile Project Management covers a variety of topics such as software designs principals, API integration, Version control and Software Development Life-Cycle approaches for methodical planning and testing. Intelligent Destination Recommendation using Natural Language Processing (NLP) adds a new level of research to the project, where the system comprehends what users input into the system and then provides suggestions of where to visit. Application of the NLP to perform destination discovery in context and the inclusion of a Logistic regression model to predict costs needs to be pre-processed efficiently and then integrated into the web application architecture Data handling gets more complicated when several APIs are integrated. The most crucial part of the project is ensuring smooth communication between various elements while preserving performance and data integrity.																					
Resources required in developing the artefact	• Database Management System: Microsoft SQL Server • Programming languages, Frameworks and Libraries: Bootstrap, HTML5, CSS3, JavaScript, PHP, Python • Backend Tools and Frameworks: XAMPP, PHP & Python • APIs and External Libraries: ○ Leaflet.js API – For mapping and location visualization ○ OpenWeatherMap API – For weather forecasting																					

	○ YouTube Data API – For displaying destination related travel videos • Tools: GitHub, Git & Visual Studio Code Data: Datasets for Sri Lankan travel destinations, hotels, cabanas, and vehicle information are imported for Kaggle and self-made datasets.	
Have you completed & submitted your ethics form?	YES	NO
If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence		

Supervisor Signature:



After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

Part 2 – List of relevant resources

Fill in this section after your project proposal has been approved by your supervisor. Use Harvard referencing (see <https://lrweb.beds.ac.uk/a-guide-to-referencing>). Modify the list below as appropriate. This list is part of Assignment 1 and will be submitted with the Project Proposal.

1. Books

- Pressman, R. S. and Maxim, B. R. (2020) *Software Engineering: A Practitioner's Approach*. 9th edn. New York: McGraw-Hill Education.
- Russell, S. and Norvig, P. (2021) *Artificial Intelligence: A Modern Approach*. 4th edn. Hoboken: Pearson Education.

2. Web Sites with relevant information

- Leaflet.js (2025) Leaflet – an open-source JavaScript library for interactive maps. Available at: <https://leafletjs.com> (Accessed: 28 October 2025).
- OpenWeatherMap (2025) OpenWeatherMap API Documentation. Available at: <https://openweathermap.org/api> (Accessed: 28 October 2025).
- Bootstrap (2025) Bootstrap: The World's Most Popular Front-End Framework. Available at: <https://getbootstrap.com> (Accessed: 28 October 2025).

3. Relevant software

- Visual Studio Code (2025) Code Editing and Development Environment. Available at: <https://code.visualstudio.com> (Accessed: 28 October 2025).

- b. *Microsoft SQL Server (2025) MSSQL Database Management System. Available at: <https://www.microsoft.com/sql-server> (Accessed: 28 October 2025).*
- c. *GitHub (2025) GitHub – Code Hosting and Version Control Platform. Available at: <https://github.com> (Accessed: 28 October 2025).*

4. *Other*

- a. *OpenAI (2025) Introduction to Natural Language Processing and Applications. Available at: <https://openai.com> (Accessed: 28 October 2025)*
- b. *Python Software Foundation (2025) Scikit-learn Documentation – Logistic Regression Models. Available at: <https://scikit-learn.org> (Accessed: 28 October 2025).*
- c. *W3Schools (2025) JavaScript, PHP, and HTML5 Tutorials. Available at: <https://www.w3schools.com> (Accessed: 28 October 2025).*
- d. *Stack Overflow (2025) Developer Discussions and API Integration Solutions. Available at: <https://stackoverflow.com> (Accessed: 28 October 2025).*